

User Interfaces (Frontend)

- Landing Page

Learn about CampusFix
Get Started / Login
- Signup

Create new account
(Student / Technician / Admin)
- Login

Authenticate user
- Student Dashboard

View & manage complaints
Submit new complaint
- Technician Dashboard

View assigned complaints
Update status
- Admin Dashboard

Manage users, complaints
Monitor system

Built with React (Vite)

Backend (Spring Boot)

REST API | Business Logic | Authentication | File Upload

1. API Layer

Handles HTTP requests and routes to controllers

POST /api/auth/signup
POST /api/auth/login
GET /api/student/complaints
POST /api/complaints
GET /api/technicians/me
PUT /api/technicians/status
GET /api/admin/complaints

2. Controllers

Validate input and call services

• AuthController
• StudentController
• TechnicianController
• AdminController
• ComplaintController
• UserController
• NotificationController

3. Services

Business logic processing

• Authenticate user
• Create / update complaint
• Assign to technician
• Update complaint status
• Send notifications
• Manage users & roles
• Handle file uploads

4. Repositories (JPA/Hibernate)

Interact with database

• UserRepository
• ComplaintRepository
• TechnicianRepository
• NotificationRepository
• ... (Spring Data JPA)

5. Models (Entities)

Map to database tables

• User
• Complaint
• Technician
• Notification
• Role / Enum

6. Response

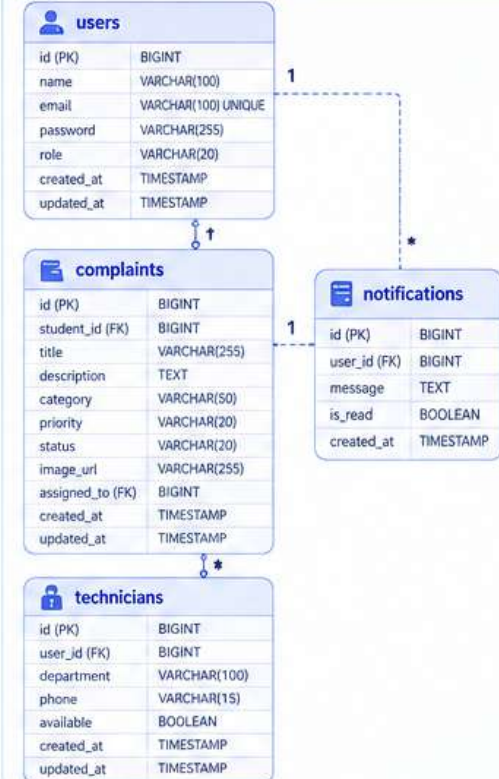
Return JSON response to frontend
(Success / Error Message)

System Flow Summary



Database (MySQL / PostgreSQL)

Main Tables (ER Diagram)



Key API Endpoints

POST	/api/auth/signup	(Register)
POST	/api/auth/login	(Login)
GET	/api/student/complaints	(My Complaints)
POST	/api/complaints	(Create Complaint)
GET	/api/technicians/my-complaints	(Assigned)
PUT	/api/technicians/complaints/{id}/status	(Update Status)
GET	/api/admin/complaints	(All Complaints)
GET	/api/admin/users	(Manage Users)

Main Features

- User Authentication**
(Student, Technician, Admin)
- Submit & Track Complaints**
(with images and categories)
- Automatic / Manual Assignment**
to Technicians
- Real-time Status Updates**
(Pending → In Progress → Resolved)
- Notifications**
(In-app for important updates)
- Admin Management**
(Users, Complaints, Technicians)

Technology Stack

- Frontend** – React (Vite)
- Backend** – Spring Boot 3+
- Database** – MySQL / PostgreSQL
- ORM** – Spring Data JPA (Hibernate)
- Security** – JWT Authentication
- Build Tool** – Maven
- Language** – Java 17+

Workflow Example

- Student submits a complaint (e.g., Wi-Fi not working)
- System saves complaint in database
- Complaint is assigned to a technician
- Technician updates status to In Progress
- Technician resolves the issue (Resolved)
- Student receives notification and views resolution



CampusFix

Campus Complaint Management System

Report • Track • Resolve • A Better Campus



Clean Campus
Safe Campus
Smart Campus

Small Issues
Make a Big Difference...!



ER Diagram (Database Tables)

CampusFix - Main Database Design

users	
id	BIGINT (PK)
name	VARCHAR(100)
email	VARCHAR(100) UNIQUE
password	VARCHAR(255)
role	VARCHAR(20)
created_at	TIMESTAMP
updated_at	TIMESTAMP

notifications	
id	BIGINT (PK)
user_id	BIGINT (FK)
message	TEXT
is_read	BOOLEAN
created_at	TIMESTAMP

complaints	
id	BIGINT (PK)
student_id	BIGINT (FK)
title	VARCHAR(255)
description	TEXT
category	VARCHAR(50)
priority	VARCHAR(20)
status	VARCHAR(20)
image_url	VARCHAR(255)
created_at	TIMESTAMP
updated_at	TIMESTAMP

technicians	
id	BIGINT (PK)
user_id	BIGINT (FK)
department	VARCHAR(100)
phone	VARCHAR(15)
available	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP

complaint_assignments	
id	BIGINT (PK)
complaint_id	BIGINT (FK)
technician_id	BIGINT (FK)
assigned_at	TIMESTAMP
status	VARCHAR(20)



API Endpoints (Sample)

POST	/api/auth/signup	(Register)
POST	/api/auth/login	(Login)
GET	/api/complaints	(All Complaints)
POST	/api/complaints	(Create Complaint)
GET	/api/technicians	(All Technicians)
GET	/api/technicians/available	(Available Technicians)
PUT	/api/complaints/{id}/status	(Update Status)
POST	/api/complaints/{id}/assign	(Assign Technician)
GET	/api/user/profile	(User Info)
GET	/api/notifications	(Notifications)



System Flow (Simple)



Example Flow: Create Complaint



Backend Project Structure (Spring Boot)

```
campusfix-backend/  
├── src/main/java/com/campusfix/  
│   ├── controller  
│   │   ├── AuthController.java  
│   │   ├── ComplaintController.java  
│   │   ├── TechnicianController.java  
│   │   ├── AdminController.java  
│   │   └── UserController.java  
│   ├── service  
│   │   ├── AuthService.java  
│   │   ├── ComplaintService.java  
│   │   ├── TechnicianService.java  
│   │   └── AdminService.java  
│   ├── repository  
│   │   ├── UserRepository.java  
│   │   ├── ComplaintRepository.java  
│   │   ├── TechnicianRepository.java  
│   │   └── NotificationRepository.java  
│   └── model  
│       ├── User.java  
│       ├── Complaint.java  
│       ├── Technician.java  
│       ├── ComplaintAssignment.java  
│       ├── Notification.java  
│       ├── LoginRequest.java  
│       ├── SignupRequest.java  
│       ├── ComplaintRequest.java  
│       └── ComplaintResponse.java  
└── CampusFixApplication.java
```

Tech Stack

- Java 17+
- Spring Boot 3+
- MySQL / PostgreSQL
- Spring Data JPA
- Spring Security (JWT)
- Maven
- React (Vite)
- REST API



Database Tables

- users
- complaints
- technicians
- complaint_assignments
- notifications



JPA Relationships (Example)

```
// Complaint.java  
@ManyToOne  
@JoinColumn(name = "student_id")  
private User student;  
  
@ManyToOne  
@JoinColumn(name = "assigned_to")  
private Technician technician;
```



DTO Example (ComplaintRequest)

```
public class ComplaintRequest {  
    private String title;  
    private String description;  
    private String category;  
    private String priority;  
}
```



CampusFix



Simple



Secure



Scalable



Built for a Better Campus

Report Today • Improve Tomorrow